



Connecting the periphery: distributive effects of new infrastructure

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ABSTRACT

Studies addressing the spatial economic development effects of infrastructure generally focus on estimating generative effects at aggregated spatial scales. However, such effects may often hide a distributive effect, which occurs when one part of a region grows faster or at the expense of another part. This paper distinguishes distributive accessibility effects and distributive centre–periphery effects. The pattern of such distributive effects is explored for the new (2003) tunnel under the Westerschelde estuary in the Netherlands, which links a central region with a peripheral region. The tunnel led to dramatic changes in accessibility since it replaced car ferries that operated at quite a distance from the tunnel. Our ex-post analyses explore whether employment and population have redistributed following the opening of the tunnel. Increased accessibility led to employment decline in the centre, and to slight growth in the periphery. In particular, the tunnel enabled a process of rationalization of employment in the non-commercial services sector. For population, we found that the new fixed link led to stronger population growth in the centre. Especially people aged 20–40 moved out of the periphery. Within the periphery, households with children relocated from areas that had become relatively less accessible to areas that had become more accessible. Hence, our study emphasizes the importance of geographical, sectoral and demographic detail in studies of the social and economic impacts of transport infrastructure.

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1. Introduction

The complex relationship between transport infrastructure investment and spatial economic development has been debated ever since the first roads, railways and canals were built. Much research has explored the link between the general level of infrastructure investment and economic development and growth (e.g. [Aschauer, 1990](#); [Fernald, 1999](#)), or examined cost-benefit analyses of individual infrastructural projects. In other words, most studies focus on estimating the generative effects of infrastructure. Generative effects refer to net (welfare) improvements that accrue to the region affected by the investment ([Rietveld and Bruinsma, 1998](#)). However, if evidence for economic development generated by infrastructure investment is provided at the national or regional scale, a distributive effect within this nation or region may often be hidden. It could well be that transport infrastructure investment leads to economic growth in one place at the expense of less growth or even a decline in another place ([Whitelegg, 1994](#); [Rietveld and Bruinsma, 1998](#); [Vickerman et al., 1999](#); [Boarnet and Haughwout, 2000](#)). Recently, interest in such distributive effects has risen as evidence suggests that the impact of transport infrastructure

development is highly localized, leading to calls for analyses of new infrastructure impacts at more geographically detailed spatial scales ([Rienstra et al., 1994](#); [Weisbrod and Treyz, 1998](#); [Rietveld and Bruinsma, 1998](#); [Banister and Berechman, 2000](#); [Knowles, 2006](#); [Holl, 2007a](#)). Only such geographically detailed studies are able to reveal distributive patterns since at the aggregated scale these tend to be lost.

A greater understanding of distributive effects is essential given that balancing the spatial distribution of economic development resulting from infrastructure development is often a major rationale for such investment decisions ([Calderón and Servén, 2004](#); [López et al., 2008](#)). Despite these cohesion-oriented intentions, in practice a distributive effect can sometimes be seen, particularly when the connection between centre and periphery is improved. Discussion in the literature centres on what is sometimes referred to as ‘the two-way road argument’ ([Preston, 2001](#)). New infrastructure between a core and peripheral region can promote outward investment just as easily as inward investment. There is little agreement on whether decentralization or polarization of economic development will take place as a result of new infrastructure ([Vickerman, 1994](#); [Bröcker and Peschel, 1988](#)) and much depends on the interplay of transportation costs and economies of scale ([Krugman, 1991](#)).

Perhaps in contrast to the abundance of literature on ex ante appraisals of effects of new infrastructure, is the comparatively limited attention for ex post studies of actual effects ([SACTRA,](#)

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1999). Holvad and Preston (2005, p. 19) state that ex post evaluations of economic impacts of major transport infrastructure that has been in operation for a period of time are necessary to 'unearth real life evidence of additional economic effects'.

Given these lacunae, this study takes a less conventional approach. It contributes to our understanding of the spatial-economic effects of new infrastructure development by (a) focusing on distributive effects instead of generative effects; (b) by examining highly localized data rather than spatially aggregated data; and, (c) by exploring ex post rather than ex ante distributive effects. Moreover, we aim to add to the knowledge of centre-periphery effects. From the literature we distill two main distributive effects and explore their appearance by means of an ex post, localized case study of new infrastructure connecting a central and a peripheral region.

The case presents an almost 'natural experiment' – the construction of the 6 km long tunnel underneath the Westerschelde estuary which, since it was opened in 2003, links the central part of the province of Zeeland in the Southwest of the Netherlands with the more peripheral part of Zeeland called Zeeuws-Vlaanderen, which borders Flanders (Fig. 1). It provided the first direct road connection between Zeeuws-Vlaanderen and the rest of the Netherlands, not taking the considerable detour through Belgium into account. For several reasons this case is of wider interest, due to the great changes to accessibility and centrality resulting from the building of the tunnel, in particular since it replaced ferries that provided connections at quite a distance from the tunnel. This implies that changes in accessibility are also strongly connected to changes in reliability, which is of increasing importance (Hilbers et al., 2004). Also, the isolated geographic location of the case study region due to its location in a river delta and as a border region limits the influence of external developments, even though the region is located in the economic core of Europe.

This paper proceeds as follows. In Section 2, we further explore the literature on distributive effects and present different patterns of distributive effects that may occur in theory. Whether such patterns occur in practice is explored for our case study. Section 3 introduces this case and Section 4 presents our research approach. The analyses follow in Section 5. Conclusions and a discussion can be found in the final Section 6.

2. Distributive effects

Distributive effects result from either accessibility effects or because of centre-periphery effects. Infrastructure investments do not lead to homogenous accessibility effects in a region. Instead, accessibility changes are highly localized, depending on the network position of places. We may theoretically expect distributive effects due to accessibility changes as the impact of new infrastructure in one place is more profound than in another. A distributive effect results when these differences in the development of accessibility of places translate into growth in one place and less growth or even decline in another. In contrast, centre-periphery effects do not depend on the actual change in accessibility, but rather relate to the integration of formerly separate markets (for labour, for retail etc.) in their entirety. When the centre grows faster than the periphery or at the expense of the periphery, or vice versa, then a distributive effect results.

2.1. Distributive accessibility effects

New transport investments and in its wake accessibility-changes may influence spatial-economic dynamics through several complex mechanisms (Rietveld and Bruinsma, 1998; Preston, 2001; Weisbrod, 2008). New transport infrastructure generally

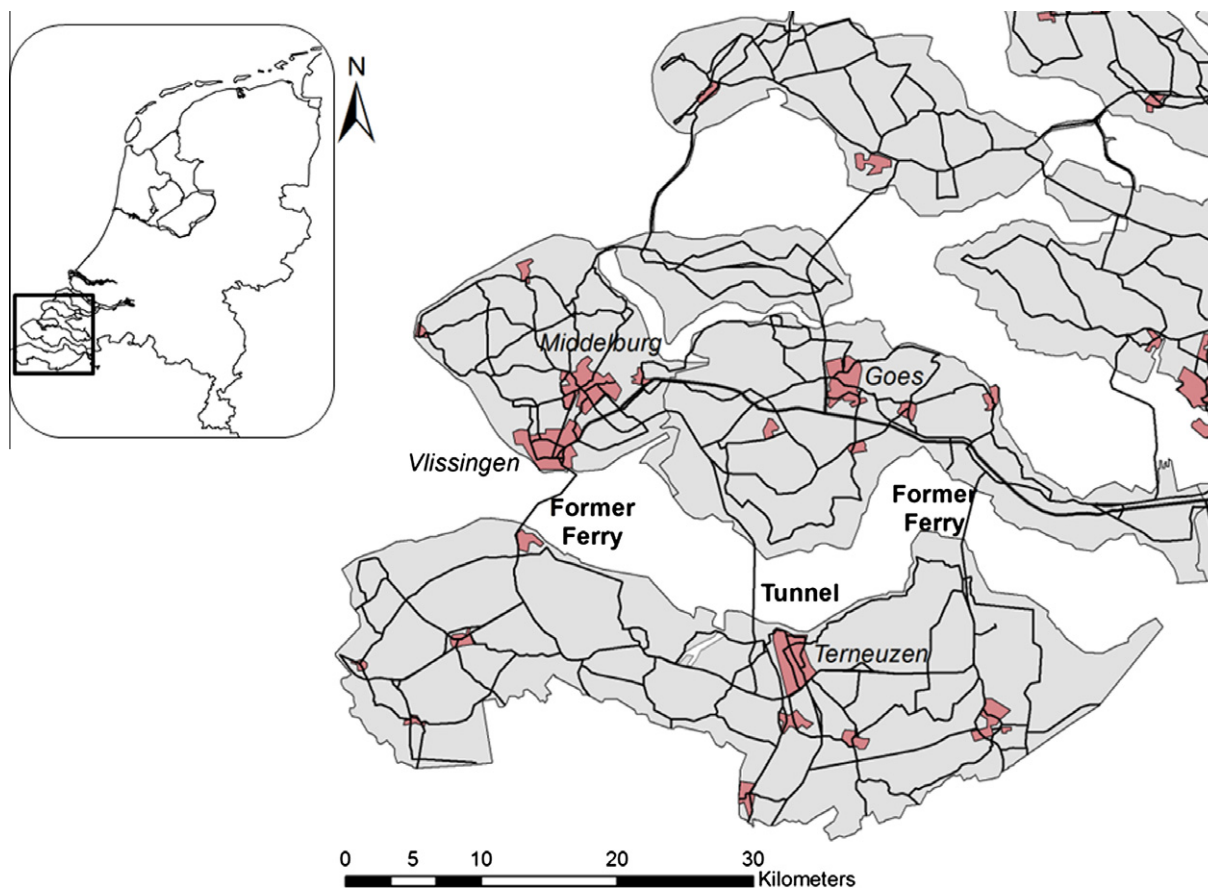


Fig. 1. The Westerscheldetunnel replacing two former car ferries.

has direct benefits in the form of travel time savings and decreased transport costs for firms. Lower transportation costs, certainly in combination with an increase in reliability, may also enable a reorganization of the logistical process within and between firms, for instance as possibilities for outsourcing, reduced input ordering times, improved output delivery and customer service are enhanced (see Bryan et al., 1997). Paradoxically, such a decrease in transportation costs may actually make transportation more important as it allows for new modes of production that depend on a high transportation input (Preston, 2001). Lower transport costs also imply an enlargement of the regional labour market and as such allows for a better match between labour demand and supply. Firms will be able to recruit their personnel from a larger pool of skilled labour. Similarly, new transport infrastructure and improved accessibility will enlarge a firm's market area.

However, the importance of these direct benefits of new transport investment are also contested. Transport costs account for only 5 to 12% of production costs, depending on the transport and interaction intensity of the industrial sector involved (Diamond and Spence, 1989) and their relative importance is on the wane compared to non-material flows. Another reason is that in advanced economies the direct benefits are often only marginal as road and rail infrastructure networks have already matured making good accessibility almost ubiquitous. New additions generally do not lead to major shifts in accessibility, either in major travel time savings or transport cost reductions (Vickerman et al., 1999; Rietveld and Nijkamp, 2000), perhaps with the exception of new types of advanced transportation infrastructure such as high-speed trains (Vickerman, 1995) or when connecting peripheral regions such as islands or enclaves. It appears that in advanced economies new transport infrastructure is not the key driver of economic growth but rather plays a supporting role, as growth is dependent on a range of other economic, investment, political and institutional conditions (Banister and Berechman, 2000; Sharp, 1980; Hart, 1993; Rietveld and Nijkamp, 2000). This would also explain the rather limited effect of changes in accessibility on shifts in employment generally found by empirical analyses (Dodgson, 1974; Nelson et al., 1994; Linneker and Spence, 1996; Bruinsma et al., 1997; Rietveld and Nijkamp, 2000; Holvad and Preston, 2005; Pugh and Fairburn, 2008).

The prime concern of this paper, however, is not economic growth but its spatial distribution at the local and intra-regional scale. According to Banister and Berechman (2000), it is the lack of one of the three sets of conditions for growth – economic, investment, and the political and institutional conditions – that will result in a distributive effect rather than a generative effect as this will change the relative attractiveness of particular places. However, one could assume that within one and the same region

differences in economic, investment, political and institutional conditions are much less pronounced than between regions. This is likely to enhance the relative importance of changes in accessibility at the intra-regional scale in explaining spatial-economic development.

In assessing the accessibility impact of new infrastructure it is necessary to consider the effect on different industry sectors. The transport intensity of production processes, the type of interaction (e.g. face-to-face contact), the labour market size, and the nature of the traded goods varies considerably between firms in different industrial sectors. Effects appear to be stronger in the distribution and services sectors (Diamond and Spence, 1989; Kolko, 1999; Ferndal, 1999). This means that we can expect that changes in accessibility will have a different impact on different sectors (Rietveld and Nijkamp, 2000; Holl, 2007b; Weisbrod, 2008). Nevertheless, studies distinguishing accessibility impacts for different sectors are rather scarce. One of the interesting exceptions is provided in Holl (2004a, 2004b), who studied the impact of motorway building programmes in Spain and Portugal, and found that certain industrial sectors were much more sensitive to being located close to new motorways than others.

Generally, distributive accessibility effects either occur when the part of a region that becomes more accessible grows while the part of the region that does not gain in accessibility grows less or declines, or vice versa (Fig. 2).

2.2. Distributive core-periphery effects

When a more central region is connected to a relatively more peripheral region, this may result in firms in the more peripheral region being able to sell their products in the core region better, leading to a significant increase in their market. It also implies that these firms will come into competition with firms in the core region, which often produce more advanced products. So, while firms in the peripheral region were previously protected from this fierce competition by their remoteness, the construction of new infrastructure may lead to a loss of regional monopoly. It is unclear whether it is the core or the periphery that ultimately benefits from this competition. The New Economic Geography approach states that the outcome depends on the balance between centripetal and centrifugal forces (Krugman, 1991; Fujita et al., 1999) and transportation costs are a central element in determining this balance. NEG models predict a reorganization of industrial location that follows an inverted U-shaped relationship between transport costs and agglomeration. In the case of high transport costs, firms need to supply their products close to consumers and therefore will spread over the territory. New infrastructure and hence a reduction in transport costs mean that firms are able to serve

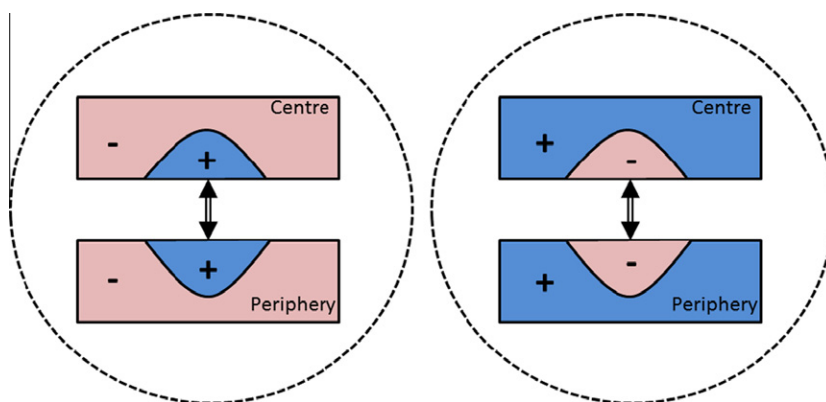


Fig. 2. Distributive accessibility effect.

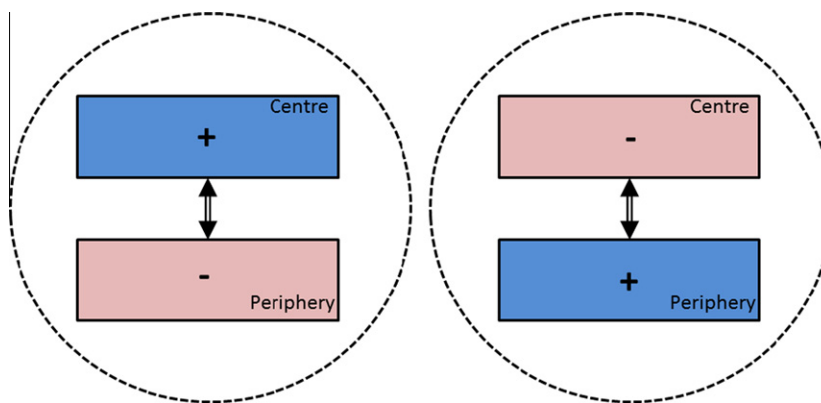


Fig. 3. Distributive centre-periphery effect.

markets from a greater distance. It is then favourable for the firms to be concentrated in a few places where it can benefit from the presence of agglomeration economies.

Graphically, a distributive centre-periphery effect occurs when the centre grows at the expense of less growth or a decline in the periphery, or vice versa (Fig. 3).

Obviously, one can imagine a variety of combinations of distributive accessibility effects and distributive centre-periphery effects. In the analyses that follow, we will explore whether such distributive effects occur, and if so, what pattern of distributive effects can be distinguished. We explore such effects by explaining several growth rates: of employment, also subdivided into several sectors and of population, which is also subdivided into several age-classes. First, we present our case study in more detail (Section 3), as well as our research approach (Section 4).

3. Case: The Westerscheldetunnel

The Rhine, Scheldt and Maas rivers flow into the sea in the delta-region in the Southwest of the Netherlands. Following a devastating flooding in 1953, the government started to strengthen sea defences in the delta, amongst others by closing off estuary-mouths, which in its wake meant that the previously isolated islands in this Delta-region became much better connected to the Dutch mainland. The Westerschelde estuary-mouth, however, was not closed off from the sea because of its importance as an entrance to the ports of Vlissingen and Terneuzen in the Netherlands and Ghent and the large port of Antwerp in Belgium. Consequently, for many decades car ferries were the only connection between Zeeuws-Vlaanderen and the central part of Zeeland and Dutch mainland, apart from the considerable detour through Belgium via Antwerp. Crossing the Westerschelde by ferry took close to half an hour, and was only possible twice hourly during daytime and if weather conditions allowed. Waiting time was not uncommon, particularly in the summer season when the Delta region attracts many tourists.

In 1995, it was decided that in the long term a tunnel would be less expensive than maintaining the car-ferries and that it could improve the connection between Zeeuws-Vlaanderen's main city (Terneuzen) and the three main cities on the other side of the Westerschelde (Vlissingen, Middelburg and Goes) by being situated somewhere equally distant from those three places. Construction of the tunnel started in December 1997 and it was opened in March 2003. At the same time, the two car ferries running between Vlissingen-Breskens and Kruijningen-Perkpolder were taken out of service. Between Vlissingen and Breskens another ferry came into service (and still operates) but is less frequent and takes only pedestrians and cyclists. The relative accessibility of places on both

sides of the river has dramatically changed since 2003. Those places that were once very centrally located have become rather peripheral, while places that were peripheral are now centrally located within the road network. In other words, the Westerscheldetunnel case provides an excellent opportunity to study accessibility effects.

It also presents an interesting case to study centre-periphery effects. Zeeuws-Vlaanderen is the prime example of a peripheral region in the Netherlands. The (formerly) poor and to-be-paid-for connections with the rest of the Netherlands made the area, with over 100,000 inhabitants, poorly accessible from the Dutch mainland. For instance, the accessibility, expressed in the number of jobs that can be reached within a 45 min time frame – was less than half the potential that could be reached from what in this study is considered the centre – Midden-Zeeland (see Fig. 4) – and a fraction of what can on average be reached from other locations in the Netherlands. However, despite being genuinely peripheral from a Dutch perspective, from a European perspective Zeeuws-Vlaanderen is not necessarily peripheral as major Flemish cities such as Ghent, Brugge and Antwerp are not that far away¹ and are occasional destinations for shopping or cultural attractions for a substantial part of the population of Zeeuws-Vlaanderen. Nonetheless, only 2% of the inhabitants of Zeeuws-Vlaanderen work in Belgium, and a considerable part of this group are Belgian nationals living in Zeeuws-Vlaanderen. This also reflects that the Dutch-Belgian border is still a strong, mostly institutional and cultural barrier, even in a unified Europe. In this study we consider Midden – Zeeland as the centre and Zeeuws-Vlaanderen as the periphery (Fig. 4).

4. Research approach

Besides the production-function approach and the transport cost approach, there is the accessibility approach to studying the impact of transport infrastructure (Jensen-Butler and Madsen, 1996). This was the obvious choice for this study, as accessibility provides the link between transportation and economic development since its change represents the impact of new infrastructure, thereby also capturing the network effects of the tunnel we study.

Our study area comprised part of the Zeeland province, more precisely the two regions bordering the Westerschelde estuary, Midden-Zeeland and Zeeuws-Vlaanderen. Our unit of analyses was the individual village or town, which, in our research area largely corresponds to the 4-digit postal code. Only the four larger

¹ From Terneuzen, the centre of Brugge can be reached in just over 1 h, the centre of Ghent in 35 min and the centre of Antwerp in about 50 min, provided there is no congestion.

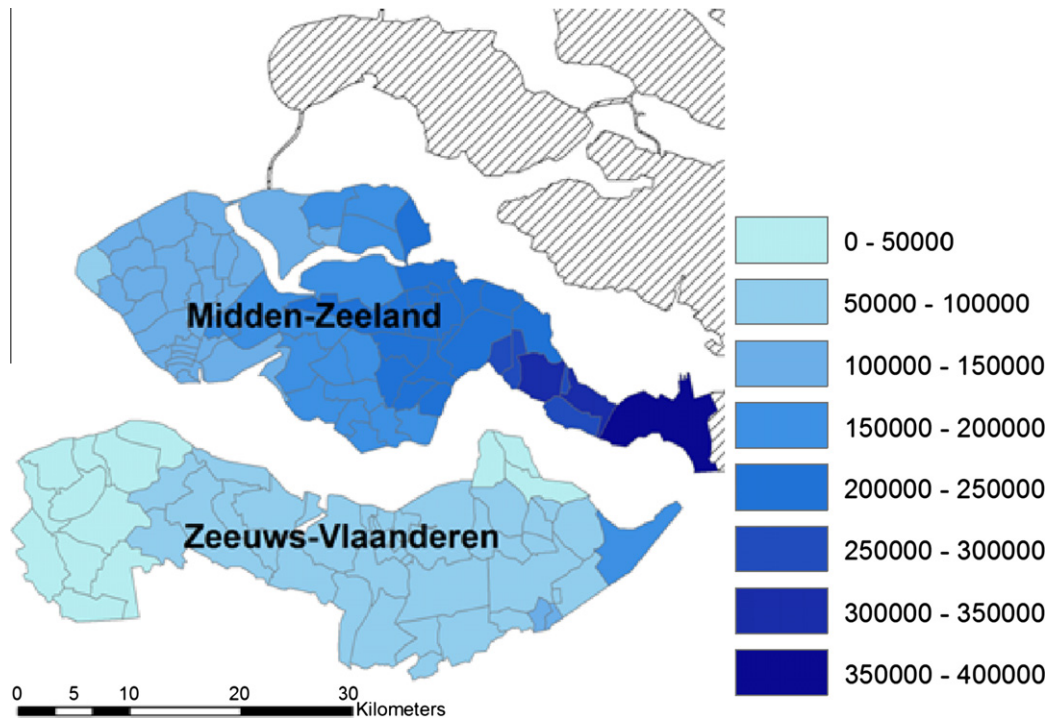


Fig. 4. Number of jobs accessible within 45 min in 2002 (before opening of the tunnel).

cities are made up of multiple 4-digit postal codes, and these were aggregated. In total, there are 102 towns and villages in the area.²

In this study, we explain the growth rates of employment and population at the local level since the opening of the tunnel (2003–2008), thereby employing OLS regression analyses. Our theoretical review highlighted the importance of studying such employment growth rates for different industrial sectors. Therefore, we developed three broad categories that together capture all employment except for agriculture and mining: commercial services, non-commercial services and manufacturing and construction. Likewise, we did not only study the total population growth rate, but also considered growth in specific age groups: under 20, aged 20–40, 40–60 and those aged over 60.

Our main interest was in the effect of three independent variables on these growth rates: the accessibility effect; the centre–periphery effect and the interaction between both. Their measurement is addressed below.

4.1. The accessibility effect

The concept of accessibility has received wide attention over the last decades and its measurement has become increasingly sophisticated (Geurs and Van Wee, 2004). We chose to use a contour measure which magnifies the variability of the accessibility effect of the tunnel and is easy to interpret and communicate. We considered the accessibility of jobs, other people and retail within a 45 min time-window, and compared the number of accessible jobs, people and retail with the Westerscheldetunnel to the situation without the tunnel but with the car ferries in service. We calculated travel times by car between pairs of 4-digit postal codes in the Netherlands, using a GIS that includes all types of roads (and average travel speeds on those roads) in the Netherlands and the main routes in Belgium. Travel times were calculated for the year 2002, just before the opening of the tunnel, as well as for the road network

in 2008. Obviously, travel times may differ not only because of the construction of the Westerscheldetunnel and the displacement of the car ferries, but also because other infrastructure changes had been made. This was controlled for, enabling us to calculate the pure effect of the tunnel. We linked our measures to local (4-digit postal code) data on employment (source: LISA), population (CBS) and retail (source: Locatus). Fig. 5 presents the percentage increase (or decrease) in accessibility caused by the Westerscheldetunnel. In our models, we used the (ln) combined average of three highly correlated growth rates – jobs (# jobs accessible without tunnel/# jobs accessible with tunnel), people (# people accessible without tunnel/# people accessible with tunnel) and retail amenities (# m² retail amenities accessible without tunnel/# m² retail amenities accessible with tunnel).

Nowadays a common alternative measure in the same family of accessibility measures includes potential accessibility measures, in which more distant opportunities provide diminishing influences by employing a distance-decay function. However, since the crossing of the Westerschelde by car ferry took about half an hour, and since it is possible to reach the other side of the Westerschelde (also by means of the detour through Belgium) within an hour, it was felt that a potential accessibility measure was less appropriate. Therefore, we employed the 45-min time window, which is not only a typical maximum distance many people are willing to travel for work or leisure on a daily basis, but also the time window which magnifies the accessibility effect of the tunnel most.

4.2. The centre–periphery effect

The centre periphery effect was simply proxied with a dummy variable, in which the periphery (Zeeuws-Vlaanderen) had the value 1 and the centre value 0.

4.3. Interaction effect

It is highly likely that combinations of accessibility and centre–periphery effects also occur, for instance because an accessibility increase in the centre has a different effect in the periphery. Therefore,

² One postal code area (#4562) was left out as the area was created and developed (construction of houses) during the period of the study, which would lead to biased growth rates.

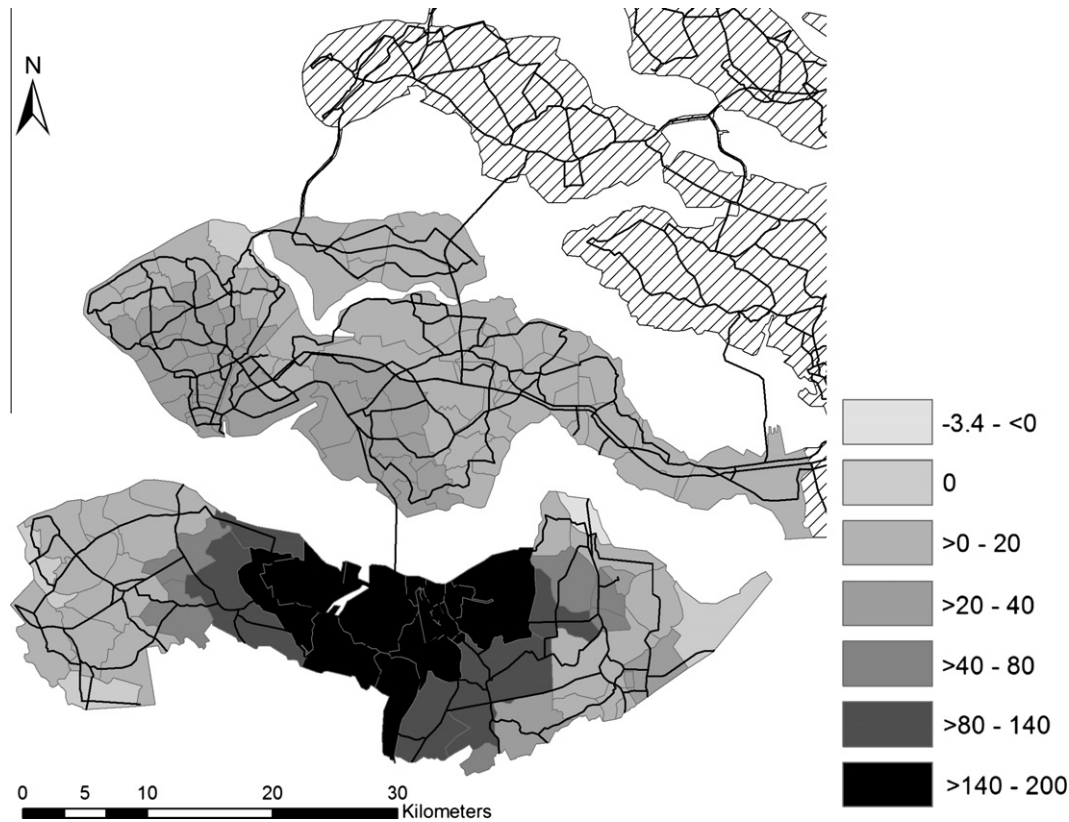


Fig. 5. The accessibility effect of the Westerscheldetunnel (average% change of jobs, population and retail that can be reached in 45 min).

we used an interaction term ‘accessibility effect’ * ‘centre-periphery effect’.

4.4. Controls

In explaining the growth rate for total employment, we included several control variables in our models. First of all, we included employment density and size, as these reflect the presence of localization and urbanization economies. When explaining total employment growth, we also took into account the composition of the local economy by including the ratio between employment in services and employment in manufacturing. That variable was left out when explaining growth rates for different sectors. We also included controls for the type of people living in the towns or villages we study. This includes their demographic composition, represented by the greying of the local population (ratio of inhabitants that are >60 years old/inhabitants ≤60 years), and we assumed that a ‘greyer’ local population tends to experience less growth. The second control variable included was the social status of the households residing in a place, an indicator provided by the Netherlands Institute for Social Research. This is a composite measure based on educational attainment, income and employment status of the population of towns and villages. A high score represents socially deprived towns and villages.

The greying of the local population and the social status of households were also included when explaining growth rates of the total population. When examining growth rates of different age groups, we left out the extent to which the local population is grey. As controls we added job accessibility within 15 min,³ as job accessibility has been a dominant force in explaining the residential location choice (Van Ham, 2002), and the presence of local

amenities per capita. The latter is measured as the net retail floor-space (m²) present in the postal code, divided by the population.

All measures are available at the detailed scale of postal codes, log transformed whenever possible, mean-centred and represent the year 2002, the year just before the tunnel opened.

5. Results

5.1. Employment growth

Table 1 presents OLS regression results for the change in total employment (excluding agriculture) at the local level. The explanatory power of the models is reasonably good. We limit our attention to our variables of interest, highlighted in bold in the table. While at first sight there is no accessibility effect (model 1) nor a centre-periphery effect (model 2), it turns out that the accessibility effect has a different impact on the centre compared to the periphery (model 3). In the centre (Midden-Zeeland), we find that a 1% increase in accessibility leads to a decrease in employment of 0.58%, whereas in the periphery, this effect does not occur (see interaction term). In fact, there is a minimal positive effect ($-0.58 + 0.61 = 0.03$) suggesting that redistribution takes place. However, this effect is only significant at the 0.1 level.

To explain this first finding, we obtained a more detailed insight by considering the growth of employment in three broad sectors: commercial services, non-commercial services and manufacturing and construction. Table 2 presents the analyses for the commercial services sector. Clearly, the Westerscheldetunnel has no distributive effects. Both the accessibility effect (model 4) and the centre-periphery effect (model 5), as well as their interaction (model 6), are far from significant.

For non-commercial services (Table 3), the picture is different. Model 7 shows that there is a negative effect for increased accessibility in employment in this sector. An increase in accessibility of

³ We would have preferred to use job accessibility within 30 min, but this is highly correlated with the centre-periphery dummy.

Table 1

Dependent: growth rate total employment 2002–2008 (ln).

	Model 1 Accessibility effect	Model 2 Centre-periphery effect	Model 3 Interaction accessibility * centre-periphery effects
Constant	0.027 (0.019)	0.039 (0.026)	−0.021 (0.043)
Employment density (ln)	−0.047 (0.042)	−0.05 (0.041)	−0.046 (0.042)
Services/manufacturing ratio (ln)	0.056 (0.021)***	0.058 (0.020)***	0.055 (0.021)**
Social status	−0.044 (0.036)	−0.042 (0.037)	−0.032 (0.037)
Greying local population (ln)	−0.352 (0.101)***	−0.329 (0.107)***	−0.326 (0.107)***
Population (ln)	0.004 (0.022)	0.002 (0.021)	−0.002 (0.022)
Accessibility effect (ln)	−0.017 (0.077)		−0.577 (0.327)*
Centre-periphery effect		−0.027 (0.041)	0.028 (0.054)
Interaction accessibility (ln)* centre-periphery			0.612 (0.333)*
R	0.493	0.496	0.523
Adjusted R ²	0.193	0.196	0.208
F	4.859***	4.945***	4.190***
N	98	98	98

All non-dummy independent variables are mean centered. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

Table 2

Dependent: growth rate employment in commercial services sector 2002–2008 (ln).

	Model 4 Accessibility effect	Model 5 Centre-periphery effect	Model 6 Interaction accessibility * centre-periphery effects
Constant	0.126 (0.035)***	0.142 (0.048)***	0.088 (0.082)
Employment density commercial services sector (ln)	−0.21 (0.064)***	−0.217 (0.057)***	−0.212 (0.065)***
Social status	−0.173 (0.067)**	−0.170 (0.067)**	−0.159 (0.069)**
Greying local population (ln)	−0.258 (0.188)	−0.224 (0.199)	−0.213 (0.2)
Population (ln)	−0.047 (0.038)	−0.048 (0.036)	−0.055 (0.039)
Accessibility effect (ln)	0.017 (0.140)		−0.527 (0.630)
Centre-periphery effect		−0.037 (0.075)	0.005 (0.102)
Interaction accessibility (ln)* centre-periphery			0.610 (0.637)
R	0.539	0.541	0.548
Adjusted R ²	0.253	0.255	0.247
F	7.717***	7.781***	5.651***
N	100	100	100

All non-dummy independent variables are mean centered. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

Table 3

Dependent: growth rate jobs in non-commercial services sector 2002–2008 (ln).

	Model 7 Accessibility effect	Model 8 Centre-periphery effect	Model 9 Interaction accessibility * centre-periphery effects
Constant	0.060 (0.058)	0.082 (0.083)	−0.272 (0.130)**
Employment density non-commercial services sector (ln)	−0.230 (0.079)***	−0.202 (0.085)**	−0.195 (0.080)**
Social status	0.031 (0.111)	0.007 (0.114)	0.085 (0.109)
Greying local population (ln)	−0.536 (0.315)*	−0.526 (0.356)	−0.660 (0.338)*
Population (ln)	0.069 (0.069)	0.038 (0.069)	0.049 (0.067)
Accessibility effect (ln)	−0.473 (0.210)**		−3.266(0.979)***
Centre-periphery effect		−0.038 (0.136)	0.371 (0.169)**
Interaction accessibility (ln)* centre-periphery			2.873 (1.005)**
R	0.436	0.382	0.511
Adjusted R ²	0.146	0.100	0.204
F	4.317***	3.146**	4.553***
N	98	98	98

All non-dummy independent variables are mean centered. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

1% leads to a decrease of employment in the public and semi-public sector of 0.47%. However, from model 9 and the significant interaction term, we can tell that this effect is stronger in the centre than in the periphery. An increase in accessibility of 1% in

the centre leads to a decrease in employment in non-commercial services of 3.27%. In the periphery, this effect is less severe. An accessibility gain of 1% translates into an employment loss of 0.4% (−3.27% + 2.87%). It is also noteworthy that, while there is

Table 4

Dependent: growth rate in jobs in manufacturing and construction 2002–2008 (ln).

	Model 10 Accessibility effect	Model 11 Centre–periphery effect	Model 12 Interaction accessibility * centre–periphery effects
Constant	–0.016 (0.054)	–0.019 (0.074)	–0.092 (0.125)
Employment density manufacturing and construction sector (ln)	–0.294 (0.065)***	–0.293(0.065)***	–0.290 (0.066)***
Social status	0.133 (0.105)	0.131 (0.105)	0.145 (0.108)
Greying local population (ln)	–0.680 (0.280)**	–0.687 (0.295)**	–0.688 (0.301)**
Accessibility effect (ln)	–0.006 (0.200)		–0.694 (0.957)
Centre–periphery effect		0.008 (0.116)	0.078 (0.159)
Interaction accessibility (ln)* centre–periphery			0.714 (0.981)
R	0.472	0.472	0.477
Adjusted R ²	0.180	0.180	0.167
F	5.264***	5.265***	3.777***
N	98	98	98

All non-dummy independent variables are mean centered. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

Table 5

Dependent: growth rate total population 2002–2008 (ln).

	Model 13 Accessibility effect	Model 14 Centre–periphery effect	Model 15 Interaction accessibility * centre–periphery effects
Constant	–0.003 (0.006)	0.015 (0.008)**	0.011 (0.014)**
Job accessibility 15" (ln)	0.010 (0.009)	0.000 (0.009)	–0.006 (0.010)
Social status	–0.006 (0.011)	–0.006 (0.011)	–0.005 (0.011)
Greying local population (ln)	–0.010 (0.034)	0.010 (0.032)	0.016 (0.032)
Amenities per capita	0.006 (0.006)	0.007 (0.006)	0.008 (0.005)
Population (ln)	–0.003 (0.007)	–0.004 (0.006)	–0.005 (0.006)
Accessibility effect (ln)	–0.011 (0.021)		–0.066 (0.101)
Centre–periphery effect		–0.041 (0.012)***	–0.045 (0.018)**
Interaction accessibility (ln)* centre–periphery			0.111 (0.101)
R	0.183	0.362	0.411
Adjusted R ²	–0.028	0.075	0.096
F	0.541	2.357**	2.334**
N	101	101	101

All non-dummy variables are mean centered. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

no general centre–periphery effect (model 8), when accessibility effects are also taken into account (model 9), we see that the periphery performs better than the centre.

In other words, we see that parts of our research area in the periphery that did not profit from an accessibility change have not experienced a decrease in employment in the non-commercial services sector but rather, keeping everything else constant, witnessed an increase. The most plausible explanation is that the integration of two formerly separate markets has led to a process of rationalization of non-commercial services. The presence of the Westerscheldetunnel makes it possible for firms to serve the whole area, centre and periphery, from one establishment, or to merge certain parts of the organization. Anecdotal evidence supports this, for instance the formerly two district water boards merged during the study period, as did the port authorities of the ports of Vlissingen and Terneuzen. A merger of the two large vocational training institutes is planned for 2012. Interestingly, the new headquarters of these large employers are concentrated on both sides of the Westerschelde. While the centre seems the most logical choice in terms of agglomeration benefits, political decisions have led to the establishment of the new headquarters of for instance the port authorities and the new vocational training schools in the periphery (Terneuzen) in an attempt to meet the widely felt feeling in the periphery that they are being neglected. Our model shows that ventilating this feeling has been beneficial.

One explanation for the fact that the part of the periphery that does not profit from an accessibility gain performs relatively well is that their remoteness protects them from the process of rationalization of service-production and from increased competition.

Finally, our discussion of distributive effects in the manufacturing and construction sector (Table 4) can be brief, since the spatial distribution of employment in this sector has not been affected by the arrival of the Westerscheldetunnel.

5.2. Population growth

Table 5 presents the analysis that explains the growth rate of the total population between 2002 and 2008. There is no indication of an accessibility effect (model 13), but there certainly is a centre–periphery effect in the sense that there is a decline in population in the periphery and not in the centre (models 14 and 15). Compared to the centre, this decline is in the range of –4% (model 14) to –4.4% (model 15).⁴ Similar to our analyses of employment, we gain more insight of the causes of this effect by studying population growth for different age groups.

The location behaviour of households with children is presented in Table 6, which presents the growth rates of the population under

⁴ $100[\exp(c) - 1]$.

Table 6

Dependent: growth rate population under 20, 2002–2008 (ln).

	Model 16 Accessibility effect	Model 17 Centre–periphery effect	Model 18 Interaction accessibility * centre–periphery effects
Constant	–0.038 (0.012)***	–0.022 (0.017)	–0.050 (0.029)*
Job accessibility 15" (ln)	0.018 (0.019)	0.014 (0.019)	–0.003 (0.021)
Social status	–0.022 (0.024)	–0.017 (0.024)	–0.013 (0.024)
Amenities per capita	0.011 (0.012)	0.010 (0.012)	0.015 (0.011)
Population (ln)	–0.020 (0.014)	–0.20 (0.014)	–0.025 (0.013)*
Accessibility effect (ln)	0.053 (0.044)		–0.317 (0.219)
Centre–periphery effect		–0.035 (0.027)	–0.035 (0.040)
Interaction accessibility (ln) * centre–periphery			0.463 (0.218)**
R	0.230	0.236	0.394
Adjusted R ²	0.003	0.006	0.092
F	1.064	1.116	2.440**
N	101	101	101

All non-dummy variables are mean centered. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

Table 7

Dependent: growth rate population aged 20–40, 2002–2008 (ln).

	Model 19 Accessibility effect	Model 20 Centre–periphery effect	Model 21 Interaction accessibility * centre–periphery effects
Constant	–0.169 (0.014)***	–0.115 (0.018)***	–0.129 (0.033)***
Job accessibility 15" (ln)	0.010 (0.022)	–0.029 (0.021)	–0.037 (0.024)
Social status	0.028 (0.028)	0.027 (0.026)	0.029 (0.027)
Amenities per capita	0.002 (0.014)	0.007 (0.013)	0.009 (0.013)
Population (ln)	0.031 (0.016)*	0.029 (0.015)*	0.027 (0.015)*
Accessibility effect (ln)	–0.079 (0.051)		–0.156 (0.248)
Centre–periphery effect		–0.124 (0.029)***	–0.123 (0.045)***
Interaction accessibility (ln) * centre–periphery			0.224 (0.247)
R	0.305	0.466	0.482
Adjusted R ²	0.045	0.176	0.175
F	1.943*	5.270***	4.030***
N	101	101	101

All non-dummy variables are mean centered. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

Table 8

Dependent: growth rate population aged 40–60, 2002–2008 (ln).

	Model 22 Accessibility effect	Model 23 Centre–periphery effect	Model 24 Interaction accessibility * centre–periphery effects
Constant	0.041 (0.008)***	0.049 (0.011)***	0.043 (0.019)**
Job accessibility 15" (ln)	0.007 (0.012)	0.002 (0.012)	–0.001 (0.014)
Social status	–0.001 (0.015)	0.000 (0.015)	0.001 (0.016)
Amenities per capita	–0.011 (0.007)	–0.011 (0.007)	–0.010 (0.007)
Population (ln)	–0.011 (0.009)	–0.011 (0.009)	–0.012 (0.009)
Accessibility effect (ln)	–0.002 (0.028)		–0.065 (0.145)
Centre–periphery effect		–0.018 (0.017)	–0.016 (0.026)
Interaction accessibility (ln) * centre–periphery			0.091 (0.145)
R	0.266	0.285	0.299
Adjusted R ²	0.022	0.033	0.021
F	1.450	1.676	1.302
N	101	101	101

All non-dummy variables are mean centered. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

20 years old. While there is no accessibility effect (model 16), nor a centre–periphery effect (model 17) in general, model 18 shows that the accessibility effect is different for the centre and the periphery.

In the periphery, a positive change in accessibility is associated with an increase in the number of children living in those places that gained in accessibility. This suggests that families with children

Table 9

Dependent: growth rate population aged 60 and over, 2002–2008 (ln).

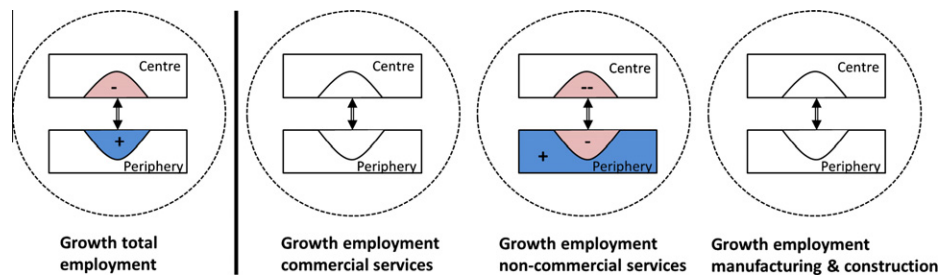
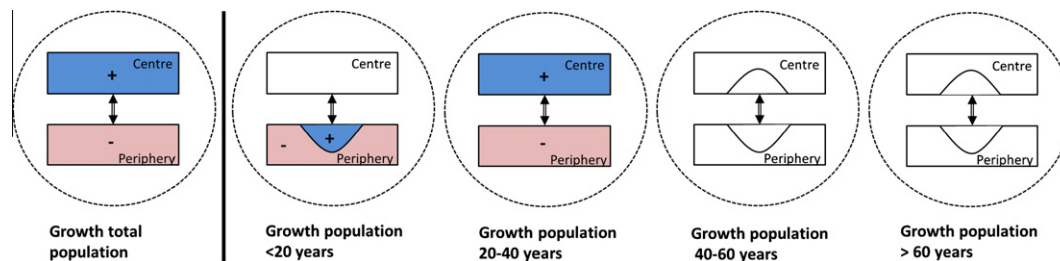
	Model 25 Accessibility effect	Model 26 Centre–periphery effect	Model 27 Interaction accessibility * centre–periphery effects
Constant	0.136 (0.011)***	0.147 (0.015)***	0.153 (0.027)***
Job accessibility 15" (ln)	0.017 (0.016)	0.008 (0.017)	0.008 (0.019)
Social status	–0.057 (0.021)***	–0.058 (0.021)***	–0.059 (0.022)***
Amenities per capita	–0.001 (0.010)	0.000 (0.010)	–0.000 (0.010)
Population (ln)	–0.000 (0.012)	0.000 (0.012)	0.000 (0.012)
Accessibility effect (ln)	–0.028 (0.039)		0.055 (0.201)
Centre–periphery effect		–0.025 (0.024)	–0.030 (0.036)
Interaction accessibility (ln) * centre–periphery			–0.062 (0.201)
R	0.322	0.331	0.332
Adjusted R ²	0.056	0.063	0.044
F	2.193*	2.337**	1.651
N	101	101	101

All non-dummy variables are mean centred. Std. error in parentheses.

* Significant at the 0.1 level.

** Significant at the 0.05 level.

*** Significant at the 0.01 level.

**Fig. 6.** Pattern of distributive effects for employment growth.**Fig. 7.** Pattern of distributive effects for population growth.

are choosing to live in the part of the periphery that has become the most accessible, at the expense of the other parts of the periphery.

Table 7 presents results for the relatively young population of places, those aged 20–40 years old. Clearly, those people are leaving the research area, but we see a rather strong centre–periphery effect (models 20 and 21), indicating that the decline was about 11.6% larger in the periphery.⁴ Younger people tend to be more mobile, and it seems that the tunnel enables them to live outside of the periphery. This may in part be a result of young people moving away for higher education (although higher education institutes are also present, albeit minimally, in the periphery), but it is also likely that younger people prefer to settle down outside of the periphery, and perhaps the arrival of the tunnel has broadened their options on the housing market.

Older age groups – those aged 40–60 (Table 8) and those aged over 60 (Table 9) – do not seem to be affected by the arrival of the Westerscheldetunnel and the resulting changes in accessibility. The positive constants indicate that both groups are growing, which indicates a further greying of the population. The eldest

age group is particularly attracted to places with a higher social standing.

6. Conclusion

Generative effects of new infrastructure development measured at a macro or meso spatial scale often hide distributive effects at the highly localized micro spatial scale. In this paper we focused our attention on such distributive effects by examining ex-post local effects of new infrastructure on employment growth and population growth. A distinction was made into two types of effects: accessibility effects and centre–periphery effects. Accessibility effects are directly linked to new infrastructure and are more localized than centre–periphery effects. They occur when a change in accessibility (gain or loss) resulting from a new infrastructure development translates into a different growth rate of employment or population. A centre–periphery effect occurs when growth rates differ between the centre and the periphery and these are linked to

the fact that the new infrastructure integrates two formerly separate markets.

The paper studied such distributive effects for the interesting case of the Westerscheldetunnel in the delta-region in the South-west of the Netherlands. This tunnel replaced two car ferries that operated at quite a distance from the site of the tunnel and resulted in a new road network where most of the central places became relatively peripheral and the once peripheral places very central. The case is not only very well suited for studying distributive effects because of the huge changes in accessibility the tunnel caused. The location of the tunnel in a delta and border region also limits external influences.

We analysed local growth rates for employment and population at the very detailed scale of individual villages, towns or medium-sized cities in the research area. As well as general growth rates we also examined the effect of new infrastructure on growth rates for different industrial sectors and for different age groups. Fig. 6 summarizes the patterns found for employment growth, and Fig. 7 summarizes our findings for population growth.

Figs. 6 and 7 highlight the main empirical findings of this study. First of all, new infrastructure development between a central and more peripheral location appears to influence the distribution of local employment and population change. Distributional effects are either linked to accessibility changes, or to centre–periphery effects. Total employment growth is linked to accessibility effects, whereas total population growth is related to the centre–periphery effects caused by the new transport infrastructure. Accessibility increases in the centre turned out negatively for employment growth, while accessibility gains in the periphery led to slightly positive rates of employment growth. However, the effect on population growth was almost the reverse. On average, the centre experienced population growth at the expense of the periphery, regardless of the local change in accessibility. One explanation for this is that the Westerscheldetunnel has integrated two previously separate housing markets. It is now more feasible to live in the centre and work in the periphery, and vice versa, but apparently the centre remains more attractive, even though house prices tend to be lower in the periphery.

A second main finding of the study is that different employment subgroups tend to react differently to the development of new transport infrastructure. We considered the effects for three broad employment sectors and we found that the new transport infrastructure had important effects on employment in the non-commercial services sector, but little effect in the commercial services and manufacturing and construction sectors. An explanation for this is that the arrival of the tunnel triggered a process of rationalization of non-commercial services, since it became possible to serve both centre and periphery from one location. It is not unlikely that such a process of rationalization had already taken place before in the commercial services sector. The manufacturing sector is dominated by some large, global firms in the port areas on both sides of the Westerschelde estuary, and for this sector, the arrival of a tunnel appears to have been of minor importance, even though they probably profited from the widening of the labour market.

As regards population growth, the tunnel influenced population growth for younger age groups. In the periphery, families with children tended to move to the part of the periphery that had become more accessible due to the tunnel. This provided them with a wider range of opportunities offered by the tunnel, such as cultural, sports and education amenities in the centre. Among the younger population (aged between 20 and 40 years), there was a general tendency to leave both the periphery and the centre of the research area and relocate to the Dutch mainland. However, this tendency was much stronger for residents in the periphery. More opportunities for education, as well as jobs, are likely to explain this pattern.

Population growth for those in older age groups (40–60 and over 60) was not affected by the development of the tunnel.

In sum, our analyses underline that the development of new transport infrastructure connecting an urban centre and periphery may lead to different development patterns across industrial sectors and age groups of the population. Even the general patterns of employment and population change can be very different. Consequently, our study emphasizes the importance of geographical, sectoral and demographic detail in future studies of the social and economic impacts of transport infrastructure.

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